

Hill Climbing Optimization and Fitness Landscape Analysis for a Discrete Unimodal Function

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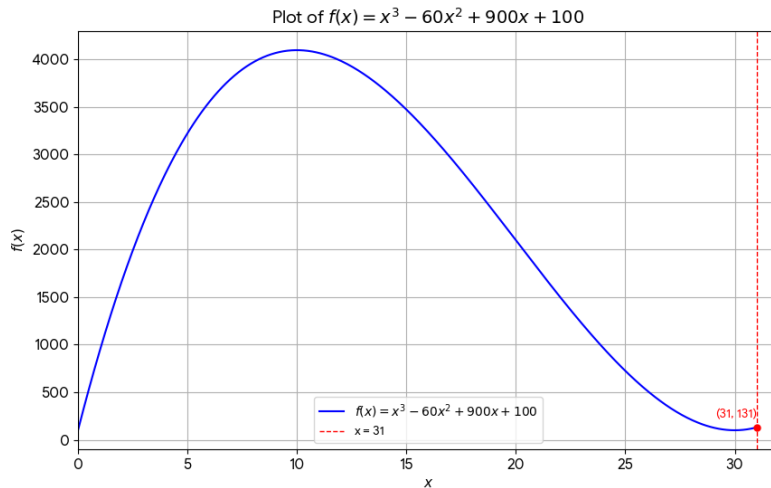
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Abstract

This report analyzes the behavior of the Hill Climbing Best Improvement and First Improvement algorithms applied to the maximization of the function $f(x) = x^3 - 60x^2 + 900x + 100$, over the discrete domain $x \in [0, 31]$. The search space was represented using 5 bit binary encoding, where the neighborhood of each candidate solution consists of all strings at Hamming distance 1. Both methods identified four local maxima, with $x=10$ as the global optimum. Best Improvement reached the global maximum from most starting points, while First Improvement more often converged to suboptimal peaks.

1 Introduction

Optimization methods are used to identify the best solution within a defined search space. This report examines Hill Climbing applied to maximizing the function $f(x) = x^3 - 60x^2 + 900x + 100$ over the interval $[0, 31]$. The search space is encoded using 5-bit binary representations, giving 32 possible integer values. A candidate's neighborhood consists of all bitstrings with Hamming distance 1. We analyze the behavior of both First Improvement and Best Improvement Hill Climbing, compare their search dynamics, and identify attraction basins leading to each local maximum.



Two Hill Climbing variants are studied: First Improvement and Best Improvement. We examine how each method explores the search space, identify the local maxima, and determine their attraction basins, meaning the starting points that converge to the same optimum. By comparing the two approaches, we evaluate how their search strategies affect convergence and the ability to reach the global maximum.

2 Method

Hill Climbing is a heuristic local search algorithm that iteratively evaluates neighboring solutions and moves to those that improve the objective function. In this study, candidate solutions are integers

$x \in [0, 31]$, encoded as 5-bit binary strings. The neighborhood of each solution consists of all bitstrings at Hamming distance 1 (obtained by flipping exactly one bit).

Two Hill Climbing variants are used:

- **First Improvement:** neighbors are checked sequentially, and the search moves immediately to the first improving neighbor. This reduces computation but may skip larger improvements.
- **Best Improvement (Steepest Ascent):** all neighbors are evaluated, and the algorithm moves to the one giving the largest improvement, ensuring the steepest ascent at each step. [2]

The search stops when no neighbor yields a higher fitness value, indicating a local maximum. To analyze the behavior of the algorithm across the full search space, the procedure is repeated for every possible starting value of x . For each starting point, the trajectory produced by both variants is tracked, the reached local maximum is identified, and the attraction basin (the set of starting points that converge to the same optimum) is recorded.

This manual simulation allows us to observe the search dynamics on the fitness landscape, compare both variants, and determine the attraction basins of all local maxima.

3 Experiments and results

The results obtained from the manual application of the Hill Climbing algorithm highlight the differences between the Best and First Improvement strategies in exploring the fitness landscape of the given function.

	Local Maximum	Values in Basin	Size
HC Best	7	6, 7, 22, 23	4
	10	0–3, 5, 8–11, 13–15, 21, 24–27, 29–31	20
	12	4, 12, 28	3
	16	16–20	5
HC First	7	6, 7, 15, 23, 31	4
	10	3, 9–11, 13–14, 21, 25–30	14
	12	4, 8, 12, 20, 24, 28	6
	16	0–2, 16–19	8

Table 1: Attraction Basins for Hill Climbing Variants

The function exhibits four local maximum points, 7, 10, 12, 16, with $x = 10$ being the global maximum. Under the Best Improvement strategy, we can see that most initial points converge to the global maximum, indicating a larger basin of attraction and a more robust search behavior. However, the First Improvement variant produces smaller and more fragmented basins, since the search process depends on the order in which neighbors are evaluated, often leading to premature convergence to local optima such as $x = 12$ or $x = 16$.

These results demonstrate that while Best Improvement requires evaluating all neighbors, it provides a more reliable convergence pattern compared to the more stochastic behavior of First Improvement.

4 Conclusions

When optimizing a unimodal function, the chosen Hill Climbing strategy strongly influences the quality of the final solution. The First Improvement approach quickly scans the neighborhood and accepts the first better candidate it finds, which makes it computationally lightweight and fast. However, this behavior can cause the search to stop too early, resulting in convergence to solutions that are not optimal.

In contrast, the Best Improvement method evaluates all neighboring solutions before making a move. Although this requires more evaluations per step, it provides a more systematic exploration of the search space and significantly increases the likelihood of reaching the global maximum. In our experiment, this method consistently guided the algorithm toward the best-performing region of the function.

Since evaluating the function in this setting is inexpensive, the additional computational effort of the Best Improvement strategy is justified. Therefore, for this particular problem, Best Improvement represents the more effective choice for obtaining high-quality solutions.

Resources and references

- [1] Lect. dr. Eugen Croitoru. *Genetic Algorithms*.
Available at: <https://profs.info.uaic.ro/eugen.croitoru/teaching/ga/>
- [2] GeeksforGeeks. *Hill climbing in artificial intelligence*. (2025, August 1)
Available at: <https://www.geeksforgeeks.org/artificial-intelligence/introduction-hill-climbing-artificial-intelligence/>
- [3] Medium. *Hill Climbing Algorithm: A Comprehensive Guide*. (2024, July 28). Available at: <https://medium.com/%40tahsinsoyakk/hill-climbing-algorithm-a-comprehensive-guide-46e33f1ecc02>